What leads to happiness?

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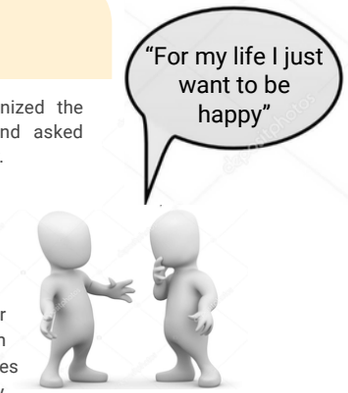
Regression Analysis – Professors Bruno Damásio, Carolina Vasconcelos, Beatriz Sousa and Carolina Shaul

Introduction



Human beings have always recognized the importance of personal happiness and asked themselves what factors contribute to it.

Understanding the determining factors of happiness can help in formulating public policies, creating more satisfying work environments, and promoting healthier lifestyles.

This work aims to estimate a linear regression model to answer the problem question. Using survey data that measures subjective well-being, we seek to identify which variables have a significant impact on happiness and the magnitude of this impact.

Main literature of the area

There have already been several studies trying to understand the key to happiness. For more than 85 years, Harvard University in the United States has been conducting the longest scientific study on happiness in history. The study follows people from childhood and analyzes each individual's personal relationships and how they face different scenarios. This study, belonging to the field of psychology, analyzes each person individually.

We take a different approach. We collected data from several countries, which we believe influence people's happiness (described in the next topic), and estimated a linear regression model. In other words, each country is treated as an observation and its columns are overall values/averages of topics of interest to the study. We are, therefore, trying to estimate the happiness of a population based on economic and social indicators in each country.

Data and Methodology

To address the economic problem presented, we used the dataset "World Happiness Report, 2005-Present" from Kaggle with 147 observations and 10 variables. We filtered the dataset to include only data from 2017 and filled the few missing values using the median of the respective columns.

Variable	Definition
Life Ladder	Average happiness value of the country
Log GDP Per Capita	Log of GDP per capita. Indicator of a country's wealth and the population's purchasing power
Social Support [0,1]	Average response to whether individuals have reliable support from relatives or friends.
Healthy Life Expectancy At Birth	Average years a newborn is expected to live in good health
Freedom To Make Life Choices [0,1]	Average response to satisfaction with personal freedom of choice.
Generosity	Residual from regressing average charitable donations on log GDP per capita
Perceptions Of Corruption [0,1]	Average response regarding the prevalence of corruption in government and businesses.
Positive Affect	Average of previous-day experiences of laughter, enjoyment, and interest.
Negative Affect	Average of previous-day experiences of worry, sadness, and anger.
Confidence In National Government	Average perception of each inhabitant's trust in their country's government

Model Analysis

We started by analyzing the correlations between the independent variables and the dependent variable. We observed that 'Life ladder' is highly correlated with 'Log GDP Per Capita', 'Social Support', and 'Healthy Life Expectancy At Birth'. Then, while investigating the correlations between the independent variables, we noticed that the same variables that had a high correlation with the dependent variable also exhibited a high correlation with each other.Given this observation, we chose to test each of these variables individually in the model, without transformations in the variables. That is, 3 models emerged, each of them excluding two of the highly correlated variables and we decided to keep the multiple linear regression model with the best result, based on the analysis of the BP test.

MLR3 No Perfect Collinearity:

The next step was to calculate the Variation Inflation Error (VIF) in order to rule out the hypothesis of variables presenting Multicollinearity. We obtained VIF < 3 for all variables, verifying there is no evidence of perfect collinearity among the predictor variables since any of the values are above 10, ranging between 1.3 and 2.7.

MLR4 Zero Conditional Mean:

Then, we had to prove that E[ui | X] = 0, based on our model. The extremely small mean of the residuals (-6.288373e-18) provides evidence that the zero conditional mean assumption is likely satisfied. Therefore, the regression model's estimators are unbiased.

MLR5 Homoskedasticity:

In order to get homoskedasticity, we had to apply some tests first (BP test, White test, and White Simplified).

BP: After applying this test we conclude that the results suggest that given the p-value of 0.0318, which is less than the conventional threshold of 0.05, and the BP statistic being greater than the Chi-Squared critical value, there is significant evidence to reject the null hypothesis. This indicates that there is heteroscedasticity in our regression model.

White: After applying this test we conclude that the results suggest that given the p-value of 0.0001864, which is less than the conventional threshold of 0.05, and the BP statistic being greater than the chi-squared critical value, there is significant evidence to reject the null hypothesis. This also indicates that there is heteroscedasticity in our regression model.

White Simplified: After applying this test we conclude that the results suggest that there is no significant evidence of heteroscedasticity in our model. The variance of the residuals does not appear to depend on the fitted values, indicating that the assumption of homoscedasticity is reasonably met.

As the BP and White tests concluded that there is significant statistical evidence to reject the null hypothesis, indicating the presence of heteroskedasticity, we will resort to the **Robust estimation**, to get a model without the presence of heteroskedasticity. We got this model:

Life Ladder = −0.674 + 0.079HealthyLifeExpectancyAtBirth + 1.874FreedomToMakeLifeChoices + 0.366Generosity − 1.127PerceptionsOfCorruption + 2.543PositiveAffect − 1.413NegativeAffect − 1.521ConfidenceInNationalGovernment

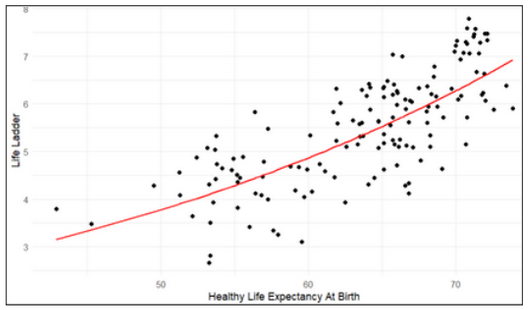
MLR6: Normality:

We can say that the assumptions of normality had been satisfied for this dataset, since we were working with 147 observations.

RESET test

A multiple regression model suffers from misspecification when it fails to properly account for the relationships between the dependent and explanatory variables. The Regression Specification Error Test (RESET) is a test used to determine whether the model has been misspecified or not. After applying it, we were able to conclude that, with a p-value = 0.03346, exists statistically significant evidence that the functional form of the model is misspecified.

Given this, we decided to check how should actually be the functional form of the model.



After testing different simple linear regression models using all of the different independent variables we were able to reach the conclusion that we should have a log-level relation between y and x.

Then, we tested another RESET test with a log-level relationship, and this time we got a p-value of 0.4197, greater than 0.05, indicating that we do not reject the null hypothesis. In other words, there is not enough evidence to conclude that the quadratic and cubic terms of the fitted values are significantly different from zero. This suggests that there is no evidence of misspecification in the new model. The quadratic and cubic terms of the fitted values do not add significant information to the model.

Conclusion and Reflection

With the final results, we are able to reach some conclusions:

- A one-unit increase in healthy life expectancy at birth is associated with a 0.016259 increase in the log of life ladder score, holding other variables constant.
- Increased freedom in decision-making choices correlates with higher levels of happiness;
- Individuals who claim to donate to charity tend to report greater life satisfaction. Even though, the coefficient for generosity is not statistically significant at the 0.05 level, indicating that it may not have a significant effect on the log of life ladder score;
- Those who perceive higher levels of corruption are likely to experience lower levels of happiness.
- Naturally, individuals who report experiencing more positive emotions "yesterday" are expected to have higher levels of happiness, whereas those reporting more negative emotions are likely to have lower levels of happiness.
- An intriguing discovery is that individuals who exhibit greater confidence in the National Government tend to report lower levels of happiness.

In conclusion, variables such as "Healthy Life Expectancy at Birth", "Freedom to Make Life Choices", "Generosity" and "Positive Affect" should have a positive impact on a person's life, and "Perceptions of Corruption", "Negative Affect" and "Confidence in National Government" should have a negative impact in person's life, although some of them have a greater impact on happiness.

Final model

The equation that explains the happiness from around the world in 2017 is:

log(Life Ladder) = 0.331698 + 0.016259HealthyLifeExpectancyAtBirth + 0.389329FreedomToMakeLifeChoices + 0.044722Generosity − 0.116766PerceptionsOfCorruption + 0.463028PositiveAffect − 0.271209NegativeAffect − 0.262061ConfidenceInNationalGovernment

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Coefficients:
                                  Estimate Std. Error t value Pr(>|t|)
(Intercept)                       0.331698   0.213612   1.553 0.122744
`Healthy Life Expectancy At Birth`  0.016259   0.002253   7.216 3.17e-11 ***
`Freedom To Make Life Choices`    0.389329   0.127791   3.047 0.002770 **
Generosity                        0.044722   0.073485   0.609 0.543792
`Perceptions Of Corruption`      -0.116766   0.076597  -1.524 0.129675
`Positive Affect`                 0.463028   0.126692   3.655 0.000364 ***
`Negative Affect`                -0.271209   0.138704  -1.955 0.052553 .
`Confidence In National Government` -0.262061   0.079145  -3.311 0.001184 **
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1206 on 139 degrees of freedom
Multiple R-squared:  0.7158,    Adjusted R-squared:  0.7015
F-statistic: 50.02 on 7 and 139 DF,  p-value: < 2.2e-16
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R-squared = 71.58% of the variability in the log of Life Ladder score is explained by the independent variables in the model.

By looking at the p-values, we can see that "Generosity" and "Perceptions of Corruption" are highly statistically significant at a 5% level, which means that they should have a strong impact on the dependent variable